THE SI SENTINEL

Weekday briefing on builders shaping super intelligence

FRIDAY, OCTOBER 23, 2026

THE MORNING PICTURE

Agents get the keys, and a key of 4 shows the stakes

A week of security scares turned into a push to lock down the credentials agents use. Guillermo Rauch summed up the certificate-authority failure at eMudhra, where one signing key was the number 4, and called for ending long-lived static credentials. 1Password launched a way for agents to sign in without seeing passwords, and David Singleton said a flaw in Muse's shared links is fixed. Aaron Levie declared security through obscurity "fully over."

Personal agents kept spreading. Grok Bot gained access to phone apps, Muse launched in Japan, and Uber gave ChatGPT Work to every employee. Microsoft published its diagnostic system in Science, Kevin Weil launched Warp Sciences with $215 million, and Elon Musk disputed a NASA panel's count of Starship refueling flights. Twenty-two of the 46 builders posted; Musk posted 15 times.

I · KEYS AND LOCKS

After a private key of 4, a push to retire passwords for machines

A certificate authority's blunder capped a heavy week in security. Vercel, 1Password and Meta answered with shorter-lived keys, approval prompts and a fix.

4
Value of one eMudhra signing key, recoverable from its public certificate

8.4%
Share of 1,000+ SaaS vendors Vercel found support secure machine sign-in

6,800+
Muse share addresses researchers found in public DNS records

Guillermo Rauch @rauchg · CEO, Vercel

Eight intermediate certificate authorities run by eMudhra, an Indian certificate authority, were revoked on October 18 after a researcher showed their private keys could be computed from the public certificates. Each key was a whole number below 256; one was 4. The security researcher Scott Helme recovered all eight in under a second, Help Net Security reported. eMudhra blamed a flawed script in an April 2025 key ceremony that made a second, weak set of keys with similar labels, and it has dropped out of the website-certificate business, Ars Technica reported. None of the eight had issued certificates to websites, and India's CERT-In cybersecurity agency issued an advisory on October 21. Rauch, at 10:48pm EDT:

> *"TL;DR of this week. Your aunt's password: "P4ssword!" (on a Post-it note in the kitchen) The stewards and guardians of the internet's encryption's password: "4" Yes, just 4. This key could be used to target and decrypt ~everything, and we're still cleaning up residual risk."*
> GUILLERMO RAUCH · @RAUCHG · 10:48PM EDT · POST

Hours earlier Vercel published "Death to long-lived static credentials," arguing that API keys that never expire are what thieves steal. It wants services to accept OpenID Connect (OIDC), in which an application presents a short-lived signed token instead. Marketplace vendors still relying on static secrets get a "Not secure" label after November 30 and face delisting in January 2027. Vercel's Malte Ubl (@cramforce) put it bluntly: "We never want to rotate API keys again." Rauch quoted the announcement at 6:38pm EDT:

> *"It's time to kill "static credentials". Attacks and breaches can happen. But imagine if thieves could only steal "self-destructing keys" to your data and tokens. The industry once came together to deliver "green locks" encryption for every site (TLS). Let's do it for OIDC!"*
>
> GUILLERMO RAUCH · @RAUCHG · 6:38PM EDT · POST

Aaron Levie @levie · CEO, Box

Responding to a post that "a lot of old software was protected by 'exploiting this would be an unreasonable amount of work,'" Levie wrote at 1:16am EDT: "Any form of security through obscurity is fully over. This barely worked as a method with humans, but just simply is over in a world of autonomous systems that will look through every crevice and try every permutation of attack possible." When Chamath Palihapitiya predicted banks would first change their terms of service to block agents "because the risk is high," Levie answered at 10:52am EDT: "This week certainly delayed things by another year."

QUICK HIT

Agent Skills goes neutral. Anthropic donated its Agent Skills format to the Linux Foundation's Agentic AI Foundation, next to the Model Context Protocol (MCP). Rauch, at 3:27pm EDT: "Super thankful to Ant for these contributions."

David Singleton @dps · Muse, Meta

The security firm Claroc reported on October 18 that it found more than 6,800 Muse share pages in a passive DNS (Domain Name System) service, because each address began with the conversation's title. Meta paused sharing Tuesday, Ars Technica reported. At 1:14pm EDT Singleton wrote that "some shared artifact URLs were discoverable via passive DNS" and that "the underlying issue is now fixed, and sharing is back on." Previously shared artifacts get new links, and Muse's bug bounty pays up to $300,000, The Verge reported.

1Password launched 1Password for Agents the same day: when an agent reaches a sign-in page, the user approves the request on their own device and 1Password fills in the password itself, so the agent never sees it. Launch partners include Muse, Grok Bot, Anthropic's Claude Duet, OpenAI's ChatGPT Work and Town's Townie. Singleton, at 12:49pm EDT: "you approve each login and your passwords stay in your vault." Alexandr Wang, Meta's chief AI officer, at 9:03pm EDT: "it'll be rolling out over the coming days—just ask your muse!"

II · THE AGENT RACE

Grok Bot reaches into phones, Muse lands in Japan, Uber signs up

Personal and workplace agents gained new reach on Thursday, and builders argued over how agents should run tools.

Elon Musk @elonmusk · CEO, SpaceX and Tesla

Grok Bot began rolling out access to Calendar, Reminders, Contacts and other phone apps on iOS and Android, with permission granted app by app, Digital Trends reported. Quoting the announcement at 9:34pm EDT, Musk wrote "Grok Bot now integrates with your phone apps." A minute later, on a user who had ordered two Garmin watches for his children with one text, he added "It's that easy." At 1:42am EDT:

> *"Grok @Bot is an incredible life & business manager. Can help you with almost anything! Like having a genius employee."*
>
> ELON MUSK · @ELONMUSK · 1:42AM EDT · POST

Alexandr Wang @alexandr_wang · Chief AI officer, Meta

Muse launched in Japan, its first market outside North America, with integrations planned for Mercari, Money Forward and Ticket Pia. Wang, at 8:41pm EDT: "konnichiwa! we are launching muse in japan today! ... (i hope you're ready for the anime memes)." Singleton added at 9:52pm EDT, "Proud of the team behind this launch." Wang also promoted Daniel Gross's $100,000 challenge to speed up Meta's 816-billion-parameter open model on Nvidia H200 chips, which runs until November 19: "$100k to the developer who can make muse spark open fastest!"

QUICK HIT

Tobi Lütke (@tobi), Shopify's chief executive, on agent tooling: "MCP vs CLI for LLMs is the wrong discussion. ... Both work incredibly well as long as they are run through a repl like environment that can persist state." At 7:41am EDT he pointed to "Excellent code mode env."

Greg Brockman @gdb · President, OpenAI

Uber said that after a 1,000-person pilot Uber is rolling out ChatGPT Work to every employee, more than 30,000 people, OpenAI said. Brockman, at 1:45pm EDT: "it's been great to work closely with Uber to roll out ChatGPT Work to every employee, bringing agentic capabilities to the whole company!"

Paul Graham @paulg · Co-founder, Y Combinator

Podium's chief executive, Eric Rea, said the company crossed $150 million in AI annual recurring revenue in two and a half

years, after taking eight years to build its first $150 million business. Graham, at 11:01am EDT:

> *"Established companies will at best adapt to AI. Newly founded startups will be AI-native from the start. This is an interesting example of a company in the middle, which (by great effort) remade itself as an AI-native company."*
>
> PAUL GRAHAM · @PAULG · 11:01AM EDT · POST

Two minutes later he called it "a good example of founder mode. A hired CEO wouldn't have the moral authority (or probably care enough) to ignore existing revenue streams and refocus the company on AI." Joe Gebbia (@jgebbia), the Airbnb co-founder, congratulated Rea.

David Heinemeier Hansson @dhh · Creator of Ruby on Rails and Omarchy

At 9:17am EDT he announced "The Basecamp plugin is now available in ChatGPT," and to a question about Claude users he replied at 9:34am EDT, "We have a full mcp too. And we've submitted the connector for review with Anthropic too."

Rauch said Vercel can now host "at-scale backends" as WebSockets reached general availability with 15 million connections a day, he wrote at 8:50am EDT. At 3:51pm EDT he described a split: "Agents are pulling software into two extremes: deep backends and just-in-time frontends," as Vercel Drop began publishing files to a live address without an account.

III · LAB AND CLINIC

A diagnosis panel in Science, a genome model and a $215 million lab

Microsoft's orchestrated models beat generalist doctors on a text-only benchmark. Arc Institute mined bacterial DNA, and a former OpenAI executive brought robots to chemistry.

80.3%	83.2%	$215M
MAI-DxO accuracy on emergency cases, versus 46.8% for physicians	Its accuracy on hard New England Journal of Medicine cases, versus 36.1%	Warp Sciences' Series A, led by Lightspeed

Mustafa Suleyman @mustafasuleyman · CEO, Microsoft AI

A paper in Science tests the MAI Diagnostic Orchestrator (MAI-DxO), which has language models act as a virtual panel of physicians that orders tests step by step while weighing cost. On 325 emergency department (ED) admissions from the MIMIC-IV-ED database it was right 80.3% of the time, against 46.8% for generalist physicians; on 304 New England Journal of Medicine teaching cases it scored 83.2% against 36.1% for doctors with reference tools. It was a controlled, text-only benchmark rather than a clinical trial, and on ED cases it spent more per case ($4,979) than the physicians ($3,400), though budget-limited settings cost less. Microsoft says the gains held with newer models such as GPT-5.5, its post says. Suleyman, at 2:13pm EDT:

> *"Our MAI Diagnostic system is now 1.7x more accurate than physicians in the ED, even when they have access to the web to look things up. It's also lower cost & does fewer medical tests to get to its diagnosis. Published in the journal Science today..."*
>
> MUSTAFA SULEYMAN · @MUSTAFASULEYMAN · 2:13PM EDT · POST

Satya Nadella (@satyanadella), Microsoft's chief executive, at 3:10pm EDT: "Important work from our team in @ScienceMagazine today, showing how our diagnostic orchestrator can reason through complex medical cases, achieving greater diagnostic accuracy while reducing testing costs."

Patrick Collison @patrickc · CEO, Stripe

Arc Institute, which Collison co-founded, posted a preprint led by Patrick Hsu that fine-tunes its Evo 2 genome model on the stretches of bacterial and archaeal DNA between genes, groups 376 million such regions into 5.14 million clusters and sends AI agents to hunt for new functional RNA (ribonucleic acid). Collison, at 8:50pm EDT: "using the latent space representation of genomes from single-cell life to find new candidate functional RNAs. I think this is cool because: (1) it uses deep learning and model interpretability to productive ends on a totally different kind of "knowledge""

Kevin Weil @kevinweil · CEO, Warp Sciences

Weil, formerly OpenAI's chief product officer and head of science, unveiled Warp Sciences, which uses AI and robotics to run preclinical drug-discovery chemistry in the United States. It raised a $215 million Series A led by Lightspeed at an $865 million valuation, TechCrunch reported. Weil wrote that the goal is "to bring frontier AI out of the browser and into the lab, and build scientific capacity here in America." Jim Fan (@DrJimFan) of Nvidia replied at 8:12pm EDT, "Congrats Kevin!!", and Levie sent emoji.

QUICK HIT

Stephanie Zhan left to start an early-stage AI fund whose backers include Sequoia and Jeff Dean. Dean, at 1:46pm EDT: "very excited to be along with you on your journey!"

Clément Delangue @ClementDelangue · CEO, Hugging Face

At 9:41am EDT Delangue criticized biology safeguards: "Frontier labs are advancing the capabilities in AI for biology but preventing the good guys (scientists, academia, non-profits, health agencies,...) to use, study and understand it with widespread but shallow guardrails that the bad guys are bypassing easily." Demis Hassabis (@demishassabis), Google DeepMind's chief executive, featured in the Wall Street Journal's Innovators issue, wrote at 3:30pm EDT that "we can collectively navigate the potential challenges of frontier models and usher in a new golden age of scientific discovery!"

IV · MACHINES AND POWER

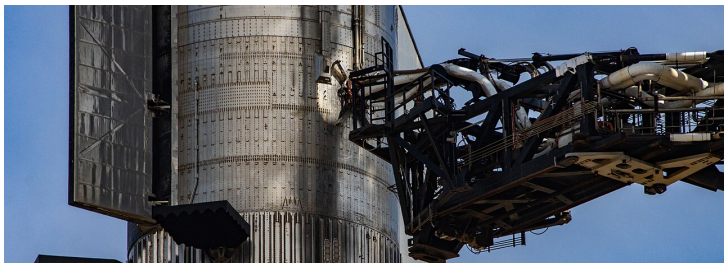
Starship's refueling math, turbines for data centers and armored dogs

Musk cut a NASA panel's estimate by two-thirds. Boom showed a turbine core, Palmer Luckey defended a dog startup, and builders marveled at a Chinese car plant.

Elon Musk @elonmusk · CEO, SpaceX and Tesla

At the quarterly meeting of NASA's Aerospace Safety Advisory Panel (ASAP), member Charlie Precourt said the latest estimate is about 22 Starship launches to fill an orbital propellant depot before a lander leaves for the Moon on Artemis IV, Ars Technica reported. When Eric Berger relayed the figure, Musk replied at 5:12pm EDT:

> *"That's way too high. Actual number will be more like 6 to 8."*
>
> ELON MUSK · @ELONMUSK · 5:12PM EDT · POST



The tower's quick-disconnect arm, which loads propellant, attached to a stacked Starship at Starbase, Texas, in April 2023.

PHOTO: JENNY HAUTMANN, "SPACE X BOOSTER FULL STACK," WIKIMEDIA COMMONS · CC BY-SA 4.0

On energy, Joe Hansen quoted Musk's earnings-call remark that Tesla is "almost sold out of all Megapacks for 2027." Musk replied at 2:13pm EDT: "We are increasing production of Megapacks as much as possible."

Paul Graham and Guillermo Rauch @paulg · @rauchg · on Boom Supersonic

Boom's founder, Blake Scholl, announced three things: its first supersonic turbine core is complete and shipping to test, its Superpower Superfactory near Denver, planned for 2 gigawatts a year of 42-megawatt gas turbines for data centers, is already making parts, and Boom is building an all-digital foundry to cast single-crystal turbine blades. The core is shared with the engine for its Overture airliner. Graham, at 4:21pm EDT: "Boom makes things happen." Rauch, at 6:07pm EDT: "It's so beautiful I'd buy it for my living room."

Palmer Luckey @PalmerLuckey · Founder, Anduril

Warhound, a startup from Ben Lamm, the Colossal Biosciences co-founder, emerged with $200 million to clone and gene-edit working dogs for the military and equip them with Anduril gear, with first puppies expected in early 2027, TechCrunch reported. Luckey disputed a WIRED headline at 12:58am EDT: "This is an awesome company that @BenLamm has started, and Anduril is very glad to have the opportunity to bring our protective Augments to military dogs, but the headline is false." To a critic who said it should be illegal, he wrote at 1:02am EDT:

> *"Which part of this do you object to? The hearing protection? The eye protection? The body armor? Or gene edits that makes them less likely to develop cancer and joint problems? There is a reason the American Humane Society is supporting this."*
>
> PALMER LUCKEY · @PALMERLUCKEY · 1:02AM EDT · POST



Caro, a military working dog with the 355th Security Forces Squadron, trains at Davis-Monthan Air Force Base, Arizona, in 2019.

PHOTO: AIRMAN 1ST CLASS KRISTINE LEGATE, U.S. AIR FORCE · PUBLIC DOMAIN

Asked how such dogs make sense in an age of autonomous warfare, he answered at 1:10am EDT, "Nothing will make you appreciate biology faster than working on autonomy." On Marathon Fusion's claim to have turned mercury into stable

gold, he wrote at 5:07pm EDT, "Nothing can replace the millennia-enduring allure of alchemy."

> **QUICK HIT**
> **John Carmack** (@ID_AA_Carmack), at 1:56pm EDT: "It is unfortunate that so much of the robotics world uses 50 hz joint targets, while the video data collection is at 60 hz. ... Easier to adjust the robot fine control rates than all the video data!"

David Heinemeier Hansson @dhh · Creator of Ruby on Rails and Omarchy

Touring China, he wrote at 1:14am EDT: "Incredible tour of the Xiaomi factory in Beijing. 91% of the construction of their new cars is automated. ... From first car in 2024 to doing 1,000 cars/day now." He added that it "completely destroys the excuse that China is thriving solely off cheap labor," and that the SU7 Ultra is "a 1,500hp car with a nicer interior than a Porsche Taycan for ~$75,000." At 1:37am EDT:

> *"I've been blown away by the energy, optimism, and hard work that has created the China I've seen on this trip. It had been seven years since I was last here, and it might as well have been forty. Invigorating to see what's possible. The West has no excuse for its managed decline."*
> DAVID HEINEMEIER HANSSON · @DHH · 1:37AM EDT · POST

Earlier, on Denmark's grid, he wrote at 7:22pm EDT that "data centers have to wait 10+ years to be connected in Denmark because wind has wrecked the grid," calling Denmark's long campaign to close Sweden's Barsebäck nuclear plant a "tragic self-own."

Also on the wire

Tim Sweeney. Fortnite returned to Apple's App Store in Australia after the Federal Court order. Epic's chief executive, at 12:21am EDT: "Epic Games Store for iOS is coming to Australia in March!" He added that Australians will be able "to install apps from the web as Scott Forstall had originally envisioned for iOS."

Clément Delangue. After Jensen Huang's keynote featured Microduck, the $399 open-source robot from Hugging Face and Pollen Robotics, Delangue said pre-orders were "going to cross 20,000+ total pre-orders probably tomorrow."

Paul Graham. A Meta filing showed Mark Zuckerberg agreed to stay in California for eight years while preparing for a proposed 5% billionaire tax. Graham, at 8:42pm EDT: "This is no more getting ready to pay the tax than buying a can of mace is getting ready to be assaulted."

Tim Sweeney on the game slump. At 4:55am EDT: "Make a great game, make it efficiently, and set the budget based on to the size of the audience that will pay for it. Technology that helps achieve that is, in my opinion, good."

Guillermo Rauch. On the data service Tako joining Vercel's AI Gateway, at 10:03pm EDT: "One of the coolest real-world examples of Generative UI."

Shopify Racing. TDS Racing expands to two cars in IMSA's Le Mans Prototype 2 class for 2027. Heinemeier Hansson, at 11:51pm EDT: "@tobi in the #11, I'm in the #29. Double the cars, double the data." Of Omarchy running on a PlayStation 5: "If it has a CPU, someone is going to get Omarchy running on it!"

Michael Dell. Dell hosted McLaren's Formula One drivers as the partnership enters its ninth year: "It's impressive to see how they use our technology to the max!"

Palmer Luckey. On a proposed Boring Company freight tunnel from the Port of Long Beach to the Inland Empire, he suggested a terminus: "Would City of Industry work?"

Takeaways

1. eMudhra's key of 4 capped a week of breaches and sped a push, led by Vercel, to replace never-expiring credentials with short-lived tokens.
2. Agents are getting sign-in rails: 1Password now lets users approve each login without the agent seeing the password.
3. Grok Bot, Muse and ChatGPT Work all widened their reach in one day, into phones, Japan and every desk at Uber.
4. Microsoft's diagnostic panel beat generalist doctors in Science, though only on a text-only benchmark.
5. Physical bets kept coming: Boom's turbines for data centers, Warhound's dogs and a disputed Starship refueling count.

On the beat

POSTED · 22

Elon Musk, David Heinemeier Hansson, Guillermo Rauch, Palmer Luckey, Tim Sweeney, Alexandr Wang, Paul Graham, Clément Delangue, Michael Dell, Aaron Levie, David Singleton, Joe Gebbia, Tobi Lütke, Mustafa Suleyman, John Carmack, Uncle Bob Martin, Jim Fan, Patrick Collison, Demis Hassabis, Satya Nadella, Jeff Dean, Greg Brockman.

SILENT · 24

Marc Andreessen, Jensen Huang, Ashok Elluswamy, Andrej Karpathy, Sam Altman, Dario Amodei, Yann LeCun, Vitalik Buterin, François Chollet, Fei-Fei Li, Andrew Ng, Michael Truell, Tom Brown, Mira Murati, Mark Zuckerberg, Noam Brown, Ilya Sutskever, Lisa Su, Sundar Pichai, Kanjun Qiu, Nat Friedman, Arthur Mensch, Guillaume Lample, Mark Chen.

Kevin Weil, the Warp Sciences chief executive and former OpenAI product and science chief, joins the watch list, which now stands at 47.

The SI Sentinel is compiled from public posts on X and published reporting. Quotes are reproduced verbatim unless marked as translated or paraphrased; emoji in quoted posts are omitted.